Improving The Adaptability of Electromyographic Signal Classification with Meta-Learning

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ABSTRACT

The scope of this study encompasses the improvement of adaptability within sEMG classification models, to encourage the feasibility of bionic prosthetics in biomedical engineering. The primary motivation behind the experiment is to address the challenges posed by the variability in sEMG signals and lack of data availability, which both make it difficult for models to quickly adapt and generalise to patient-specific gestures from only a handful of repetitions.

The methodology proposes a new model, ACEnet - a rudimentary Convolutional Neural Network. The research questions are based around the effectiveness of implementing Reptile; a MAML-style Meta-Learning architecture developed by OpenAI, in a bid to improve the model's accuracy and adaptability in a few-shot environment. Testing in a few-shot environment on unseen data answers the adaptability research question, and allows the results to translate to a real life worst case scenario. In addition, the training dataset is augmented and expanded, to test if this improves the results further.

Two datasets are used in their raw data form: the NinaPro DB2 for training & validation, consisting of sEMG signals from intact subjects, and the NinaPro DB3 for few-shot testing, which consists of sEMG signals from amputated subjects.

The findings show that data augmentation significantly improved ACEnet's accuracy when tested on an unseen subject, from 42% to 49%, indicating that more diverse datasets allow for improved adaptability to unseen data. Similarly the training accuracy improved from 65% to 70% with augmentation, falling into the same bracket as the benchmark papers discussed in section 5.3.

In the Few-Shot experiment, the highest accuracies achieved were 34% and 35%, which is similar to the primary benchmark study's results.

Despite this, the primary hypothesis of Reptile increasing the accuracy/adaptability is not supported by the results. In the few shot testing experiments, negligible results were achieved in both the non-augmented and augmented tests, showing only 5/11 and 6/11 occasions on which ACEnet achieved a higher accuracy.

In conclusion, the study contributes to filling the literature gap on Meta-Learning in sEMG classification. Despite the hypothesis failing, the study highlights the importance of preprocessing, such as Short-Term Fourier Transform, for accurate classification. The study also lays the groundwork for future studies to build upon and improve the adaptability. Ethical considerations include proper handling of data, and compliance with legal regulations regarding data privacy and usage.

1.0 Introduction

1.1 Context

Electromyography is an imperative element in developing bionic prosthetics, originating from the discipline of Cybernetics [1], but the concept stems back to 1510, when Paré crafted the first documented functional limbs [2]. Come 2013, the first relevant AI powered prosthetic was manufactured by DEKA [3], opening up the possibility for bionic prosthetics to become feasible in medical fields.

Despite the recent advances, for prosthetic systems to be feasible in today's society, they must be able to quickly adapt and generalise to the variable unique sEMG (surface electromyography) signals of individual patients. Due to many underlying factors such as environment, anatomical/physiological properties of muscles, and sensor placement, all sEMG signals are distinct, making identifying their representative features a complex task [4].

Furthermore, this classification must be achieved using only a very small number of training examples, as every patient cannot be expected to record such a vast dataset. In current research, accurate classification requires a few hundred or thousand data points at the very least [5], which is inefficient, and renders the prosthesis unusable on a wider basis. One possible solution to this is the implementation of 'Meta Learning', which allows models to adapt to unseen data with very few training examples [6].

Aztori et al. [7] demonstrate the variability in sEMG signals by recording both intact and amputated subjects signals in an identical environment. The implementation of a model that can adapt quickly when trained on vastly different data would be a great step towards improving the feasibility of bionic prosthesis in modern day biomedical engineering.

1.2 Motivation, Aims & Objectives

1.2.1 Motivation

This study is motivated by the current state of prosthetic limbs; it is simply not feasible for amputees. A recent 2022 survey found that 44% of patients abandoned their prostheses for various reasons [8], showing a necessity for improvement. This is further supported by [9] and [10], both stating the current difficulties and potential benefits surrounding bionic prosthesis.

The overarching motivation of this study is the desire to contribute to filling the knowledge gap surrounding meta-learning within the sEMG classification field, through the development of a rudimentary Convolutional Neural Network model named ‘ACEnet’ (Adaptable Classification Electromyography Network). The study aims to advance the current knowledge of sEMG classification as a whole, which following further research, allows for bionic prosthesis to become a feasible solution for amputees, solving a real world problem and benefitting multiple stakeholders, such as amputees, medical professionals, manufacturers and so on.

1.2.2 Aims

- Fill the literature gap regarding meta-learning in the field of sEMG classification
- Advance the current knowledge of sEMG classification
- Contribute to solving real world problems surrounding bionic prosthesis

1.2.3 Objectives

- Develop ACEnet, a foundation adaptable sEMG classification CNN that can be built upon in further research
- Test the effectiveness of Reptile in improving the adaptability of a CNN when dealing with unseen, small datasets in a few shot environment

- Compare the accuracy and adaptability with a typical CNN when dealing with the same dataset
- Explore the potential accuracy increase when using augmentation to produce synthetic samples and expand the small dataset
- Contribute foundational knowledge for future studies to the biomedical engineering field using meta-learning
- Provide a comprehensive analysis of the results with a discussion of future research and a conclusion

1.3 Research Questions & Hypothesis

The primary and secondary research questions of this study are as follows:

Can we successfully produce a more adaptable Neural Network model and improve the accuracy of sEMG signal classification using meta-learning?

Can data augmentation based expansion of small datasets contribute to improved accuracy of the model?

The hypothesis of this research is that the implementation of Reptile [11] Meta-Learning with a CNN model will outperform a standardly trained CNN in terms of classification accuracy, and be more adaptable to unseen, non-ideal data when tested in a few-shot environment.

1.4 Objects of Analysis

The primary object of analysis in this study is the model accuracies, but specifically the difference between the accuracies of a conventionally trained model, and ACEnet. This will be visualised, explored and compared using the tools discussed in the methods section.
The study will also look at the convergence time of the models, accuracies, loss and general patterns in the results to draw a conclusion, and ensure that the results are valid, e.g. not overfitting or displaying anomalies.

1.5 Contributions

Overall, This study proposes to:

1) employ a new Meta-Learning architecture (Reptile) to the ‘sEMG classification field’ and assess its results. Successful or not; the study will fill this literature gap.

2) Apply an augmentation method that has been certified as successful in previous literature [12], [13] to assess if dataset expansion can affect ACEnet’s accuracy & adaptability.

3) Contribute to providing the foundations for future research in meta-learning for sEMG classification through valid results.

4) Solve the current adaptability issue and handle problems surrounding small datasets/lack of data within the sEMG field.

1.6 Chapters Outline

In the subsequent chapters, analysis of the current state of research in the field takes place, followed by a discussion of the tools and methods used to achieve this study’s goal and carry out the experiment with a thorough analysis and conclusion.

2.0 Literature Review

Research in the field of sEMG signal classification is currently rather niche. A handful of studies have explored different approaches and methodologies, however the challenge of adaptability still remains a common factor.

2.1 sEMG Classification with CNN’s

Convolutional Neural Networks (CNN’s) have proven their superiority as an effective architecture for classification tasks [14], and served as a basis for the development of improved models [15]. CNNs consistently achieve classification accuracies that surpass those of other models [16], [17].

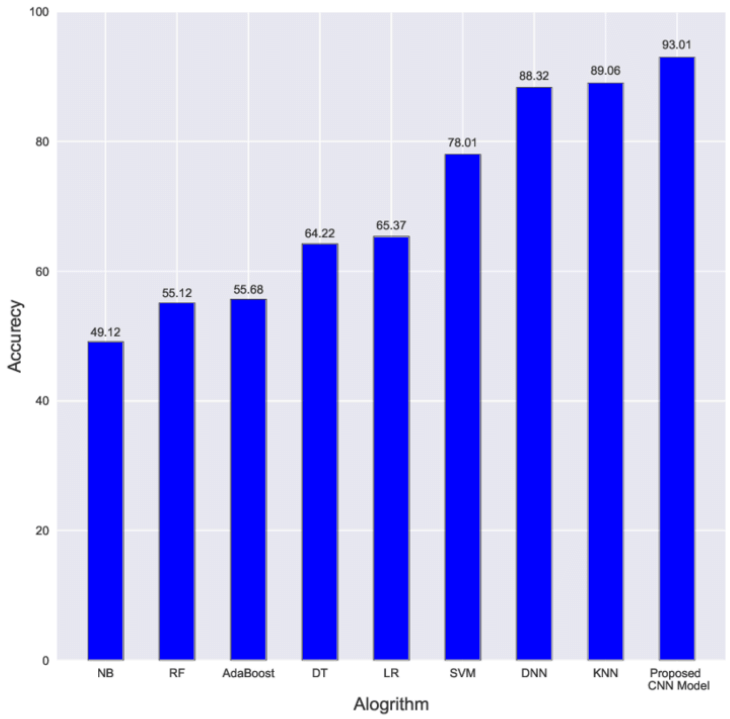


Fig. 1 - comparison chart from [17], a study that classifies network intrusions

A notable study by Erozen et al. [18] makes use of two CNN's, Slow Fusion and Inception, and tests both on a small dataset consisting of 36 participants, with the first achieving 80.88% and the latter reaching 82.64%. Erozen then created a hybrid model of the two, which produced an improved accuracy of 83.97% [18].

Ding et al. [19] also used a CNN model (C-B1PB2) to classify sEMG signals. They employed parallel multiple-scale convolution with larger kernel filters, and used the NinaPro DB2 for evaluation. An 83.79% accuracy was achieved.

In both papers, when a hybrid method or approach was introduced, better accuracies were achieved than the standard models in all cases, showing that research must continue to explore CNN's in conjunction with new methods, just as this experiment aims to do.

A critique regarding the two papers above is that they both performed on very small datasets, and it is not specified if the results were tested on unseen data or not, making it plausible that

unreliable results were produced due to overfitting [20, p20]. This would fail to adapt in a real scenario.

Expanding on this point, further research indicates that combining CNN's with RNN's (recurrent neural networks) can provide even higher accuracies. Yi Hu et al. [21] found this to be the case due to the RNN's ability to capture temporal properties of the signal well, whereas a CNN focuses solely on the spatial information [21], achieving accuracy increases of 9.2% and 3.5%.

The type of RNN used in [21] was an LSTM (long short-term memory), which allows the network to learn a more defining feature space by remembering its previous states [22]. LSTM's use for sEMG classification has proven successful in previous literature, specifically when combined with another architecture [23].

From the above papers, a clear gap in the literature surrounding CNN's being used for sEMG classification can be observed, in that combining CNN's with other adaptations can produce more accurate results.

This study proposes to introduce a new adaptation (Reptile Meta Learning) to the field, and strictly test its performance on unseen data, dealing with both of the problems discussed above that the papers failed to explore.

2.2 Meta-Learning Concepts for Improved CNN Adaptability

Meta-Learning is a contemporary method of allowing networks to learn their initial parameters from a multitude of structurally similar data [24]. This helps the model to generalise and adapt to new data faster. Previous literature has shown remarkable results when implementing meta learning methods [25], [26].

One such method is the 'Model-Agnostic' style of Meta Learning (MAML) [26]. The MAML approach allows for the implementation of 'few-shot-learning' [27] which is ideal for achieving the aims and objectives of this study, due to its ability to learn from very little training data thanks to its initial weight initialisation [28, p14].

Other methods within the meta-learning discipline were also considered and evaluated to achieve this study's hypothesis. One being Stacked-Learning, which focuses on combining the predictions from multiple learning models into a final predictor. A paper by Shen et al. [29] made use of this method for sEMG classification, which achieved at best an accuracy of 76.02%. However, Shen et al. [29] do not answer the adaptability question, as the model was not tested on unseen data from new subjects, and it is also complex and computationally expensive due to using multiple models, making it unsuitable for this study's 'fast adapting' requirements.

MAML architectures however, do solve the adaptability problem. This is proven extensively by Finn et al. [26] testing their model on a tiny dataset with MAML; it showed continual improvement with no signs of overfitting [26, p6, fig.3]. To validate the results, they also tested the model on an unseen few-shot dataset, and achieved 48.70%. [26] presents solid evidence that MAML is suitable for this study, however further research shows an even better contender.

Nichol et al. [11] introduced the MAML-based Reptile, which they discovered performed just as well as its predecessor [7, p9], whilst requiring less computation due to using only first-order derivatives to make updates (Nichol et al. 2018) [11]. It is also notably easier to implement than MAML, due to its simplicity, and integration with other libraries such as Tensorflow and Pytorch [30]; tools that will be used in this experiment.

[11] also shows successful results when testing Reptile in a few-shot environment, which also aligns with the intentions of this study.
Building upon this decision further, Moosman [31] found that in comparison to other meta-learning algorithms, Reptile reduced the training time and epochs by 80%, and halved the amount of the training data required [31].

It is notable in [21] that diverse training datasets with a large number of classes are used; Omniglot and Mini-Imagenet. This is also seen in various previous implementations, [32], [33], indicating that Reptile performs well on diverse datasets. This study will verify or falsify this by testing on a small dataset and an expanded one.

To compare the papers, [26] is much more empirical, whereas [11] is primarily theoretical, as it introduces a new method. Despite this, Nichol et al. [11] still test on variations of datasets [11, p9] and discuss the comparison of their method to MAML, which in turn provides trustworthy results.

Both papers are in agreement that MAML style implementations are highly beneficial for use with CNN's and small datasets, due to their ability to generalise from small amounts of examples.

To conclude, the above research shows Reptile to be the most suitable meta-learning approach to apply to this study, as it answers the research questions and aims.

2.3 Dataset Expansion Through Augmentation

Improvements in computer vision have often been down to the "availability of large scale datasets", however "obtaining such comprehensive datasets remains a challenge" [34]. The sheer volume of data required is becoming increasingly unmanageable, as our models become larger, and attempt to tackle more intricate tasks. This study is an example of this issue. Image classification is no problem, as there are expansive image datasets available, but for complex unique data, such as sEMG signals, recording and building datasets is challenging and unresourceful.

The previous section found that Reptile performs well on more diverse datasets, however the current electromyography datasets lack such diversity, and therefore a method to produce more data may improve results.
One method of achieving this is through data augmentation; creating synthetic samples from altered versions of the original data.

This has proven possible throughout previous literature. One notable study by Coelho et al. made use of a WGAN to produce 200 synthetic sEMG signals [35]. The addition of these signals provided a 4.07% accuracy increase when deployed using a CNN. There have also been other successful use cases, such as [36] which utilised a GAN to produce tabular data.

Another method known as ‘Window Warping’ was demonstrated by Iwana et al. [13] in a fantastically comprehensive paper. They tested various methods on a multitude of datasets, and found Window Warping to produce the best results overall.

The paper’s both take a very different approach in terms of methodology. Iwana et al. [13] provided a much more thorough investigation, and produced a solid final result that can be validated by the sheer amount of experimentation and comparison seen in the study. Coelho et al. [35] compared their WGAN to a few standard augmentation methods (warping and scaling), and found that it outperformed both, however this is not comprehensive enough, and the methods are too dissimilar to be comparable.

The papers do however both prove the hypothesis that data augmentation can produce a significant increase for classification accuracy. They also both address the problems surrounding the expansion of small datasets, specifically regarding time-series data, validating its use in this study.

Unfortunately, in [35] the researchers do not discuss the computational cost of their method. Wider reading into the topic found that standard GAN’s can cost around 3x the standard compute time than traditional augmentation methods, and do not perform much better than them (Perez 2017) [37, p7]. This statement is somewhat troublesome, as it conflicts with the comparison results in [35], which show a reasonable improvement, however the high computational cost of using a GAN/WGAN for this task coupled with the resources and time required to optimise and train it, makes it an unfeasible choice for this experiment.

Iwana et al. [13] on the other hand do discuss their computational efficiency in sections 7.3 and 7.5.3 [13], and produce a valid recommendation of a lightweight augmentation method (WW) that showed a fantastic improvement. This comprehensiveness makes it a trustworthy choice for this study..

2.4 MAML for sEMG Classification

Despite the large gap in the current literature regarding sEMG classification with meta-learned models, one recent paper from May 2022 by Fan et al. [38] demonstrated

success when employing a MAML-trained model on sEMG data. One of the paper's goals was to improve the model's classification accuracy when employed on unseen data, to make it fast and adaptive, which matches this study's objectives. The researchers did use multiple unique pre-processing methods, however they also compared the results to a test without these implementations, along with varying factors such as different batch sizes and so on, making it an excellent and comprehensive paper to review. From the current literature, it proves itself as the most similar and useful piece of work for this study, and will contribute to this experiment's choices.

The tables in [38] show the top accuracy achieved for each input form; and then the mean accuracy is taken, resulting in "82.50%, 81.00%, and 80.91% (DB-a, DB-b, and DB-c)" [38]. This is a reliable method for scoring the final accuracy/adaptability, as it allows room for variability in the outputs.

Fan et al. [38] included some helpful 'Ablation Studies' in their paper, which concluded that the STFT resulted in a large improvement when classifying the gestures of new subjects. Previous researchers agree that a STFT is an excellent choice for feature extraction [39], and can in turn provide better results, making it a consideration for further research.

2.5 Research Gap & Current Literature Findings

It can be deduced that CNN's are a state-of-the-art method for sEMG classification, however the research in the area as a whole is still limited and challenged by the lack of data and explorative analysis.

It is also clear that additional methods such as meta-learning, augmentation and hybrid models show promise for solving the research questions when used with a CNN, as each paper above showed improvements in accuracy and better generalisation to new data. These findings lay the foundation for developing a more robust, adaptable model for bionic prosthesis as a whole, advancing the field.

The literature review also discovered from [11] [33] [32] [31], that the nature of Reptile's learning method allows it to perform well on more diverse datasets. As a result of this

discovery, data augmentation will be applied to increase the diversity of this study's dataset, and the results will be analysed.

The existing literature lacks comprehensive exploration of Reptile's use in sEMG classification. However, the paper previously mentioned, by Fan et al. [38] demonstrates improved classification results when implementing MAML for this task. It is important to note that Reptile is not a derivative of MAML, but rather first-order MAML (FOMAML), and therefore may produce variable results, due to the differences in architecture when employed in this study.

This study follows a similar hypothesis and methodology of [38], and will mimic some parameters, but instead uses a stricter experimental setting, with different model-training and data preprocessing methods taken from [13] & [11] in an attempt to hone in on the adaptability to new patients.

To summarise, this review establishes CNN's as the state-of-the art model for classification tasks. It continues on to reveal a research gap in the field of sEMG classification, in which Reptile lacks investigation. Despite the minimal research, the papers discussed [38], [11], [26] provide reasonable evidence that incorporating Reptile into this context would present promising results. The review also identifies Window Warping, a proven successful method of expanding a dataset, which can be utilised in this study.
Finally, it delves further into the MAML architecture that Reptile is built upon, to explore possible outcomes and previous results of similar studies.
Collectively, this research provides solid foundations and motivation for the proposed experiment, by exploring literature gaps and the best possible tools & methods to fill them.

3.0 Methodology

3.1 Research Strategy

A quantitative approach was chosen for this study, most importantly because it allows for testing the relationships and patterns between variables, and deals with statistical analysis

(Creswell, 2009) [40, p137]. This study will evaluate the performance metrics of the models, allowing for a conclusion that can answer both research questions directly.

Quantitative research now derives from postpositivism which is based on testing, and building evidence to reject or support the hypothesis (M. Crotty, 1998) [41]; such an approach is useful for answering this study's research questions, as it's field is currently unexplored, and therefore the potential outcome is currently unknown. Exploring a new field means that this study will be foundational research for future projects, so it is vital that other researchers can reproduce, verify, and work with the study's findings [42]. The experiment itself will be strict for this reason, and performed in a tough 'worst case scenario', to ensure verifiability within the results. As discussed by Hinsen [43], human intervention makes it challenging to replicate results, and therefore the starting grounds must be as solid as possible.

Quantitative results are objective, precise and more reproducible than qualitative [44], which gives a solid answer to the research questions, as the metric comparison will provide a simple yes or no.

Fig. 2 depicts the full project roadmap. Each section of this chapter corresponds to a part of the roadmap. The figure and sections cover both the ACEnet and Standard comparison CNN models.

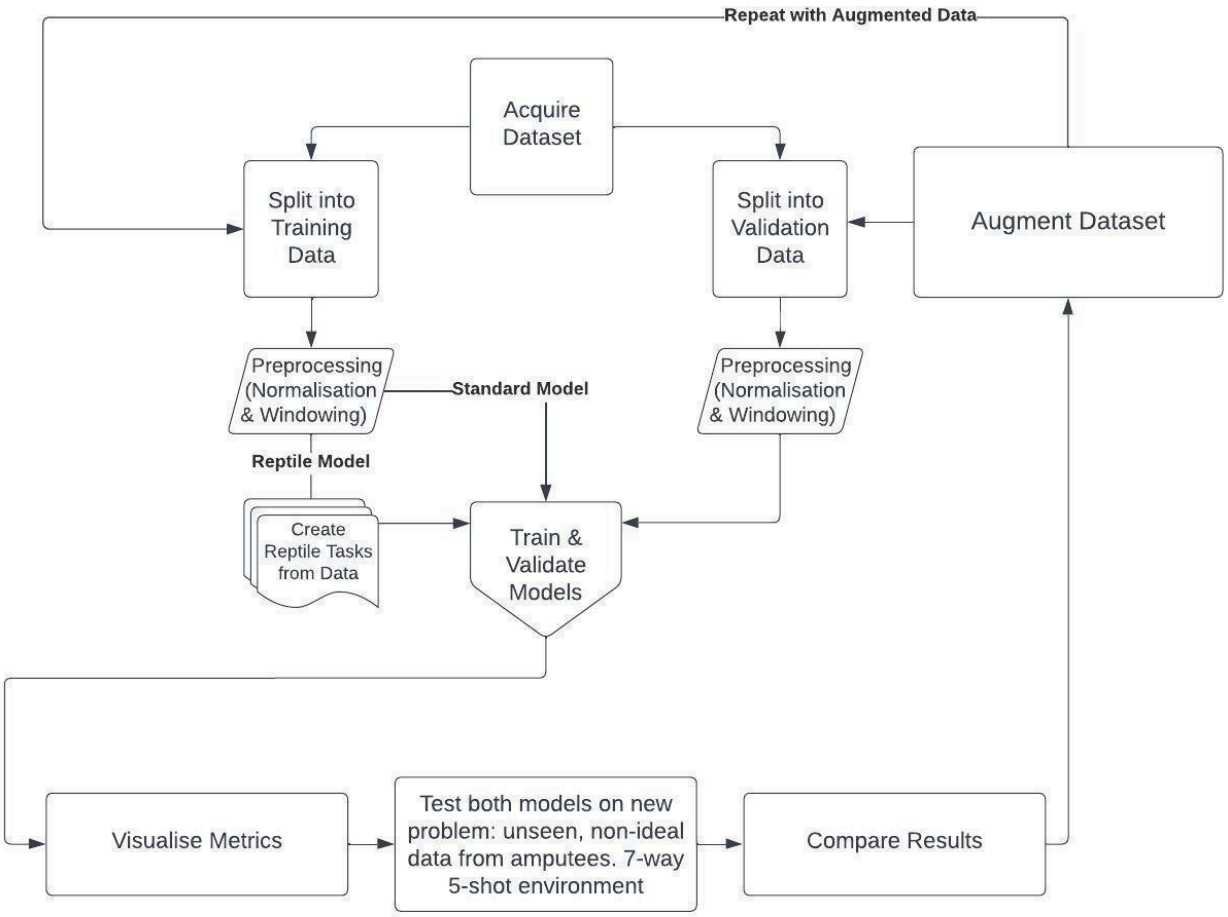


Fig. 2 Roadmap for the study

3.2 Tools and Libraries

3.2.1 Python

The Python programming language will be the overarching tool used to run this experiment and achieve the hypothesis, due to its inclusion of extensive ML libraries such as Tensorflow [45, p237], which will be used to build and analyse the models and process the data, along with other necessary libraries such as Numpy and Matplotlib.

3.2.2 Tensorflow

Due to the use of Reptile with sEMG signals being an entirely new piece of research, attacking a unique problem, the model must be built from scratch.

Tensorflow is a machine learning library that boasts a comprehensive literature and integrates flexibly with the other libraries that this study requires, making it a good choice [46], [47].

Pytorch also stood as a secondary consideration, as it is equally robust, however the original Reptile implementation was produced in Tensorflow [33], and therefore has support online that will be beneficial to combat any errors during the experiment.

3.2.3 Matplotlib

Matplotlib is highly beneficial for quantitative analysis and visualisation [48, p231-232]. It is crucial to this study to plot the accuracy increase of each model over time during training, and visualise the final results. This data will be used to make comparisons between the models, and see which one converges later, reaches higher accuracy, has the smallest loss, and so on.

These metrics will be used to draw a conclusion as to whether the experiment was a success or a failure, and to help identify anomalies that need correcting during the process.

3.3 Dataset Acquisition

Time-series datasets of sEMG signals will be acquired and used for this project; the NPDB2 [7] for training, and NPDB3 [7] for testing. Both dataset's papers clear them for research use, and the subject's data is anonymised. The NPDB3 represents realistic data obtained from real-world conditions [49, p1005] which helps to work towards achieving the study's goals by making it applicable to real life implementations. As mentioned in the literature review, a lot of the current papers do not test on non-ideal, unseen data, but rather a partition from the same dataset [18] [19] [29]. This can introduce a bias, and therefore does not translate well to a real world implementation. To fill this gap and produce valid and reliable results, this study tests on the NPDB3's unseen data.

The original paper [7] for the NPDB2 & 3 uses gesture repetitions 2 and 5 as the test data, leaving 1,3 4, and 6 as the training data [7]. These settings will be mimicked in this study as they produced successful results in the original paper.

3.3.1 NinaPro Database 2

The NPDB2 [7] has been used successfully in much of the literature seen in this study, and wider reading shows further applications to other scenarios/tasks [50], [51]. The creators make use of 12 electrodes, providing a comprehensive signal for 49 physical gestures. The time-series data is recorded at 2 kHz from a mixture of 40 subjects as seen in fig 3.

Title	Hand	Laterality	Gender	Age	Height	Weight
01	Intact	Right Handed	Male	29	187	75
02	Intact	Right Handed	Male	29	183	75
03	Intact	Right Handed	Male	31	174	69
04	Intact	Left Handed	Female	30	154	50
05	Intact	Right Handed	Male	25	175	70
06	Intact	Right Handed	Male	35	172	79
07	Intact	Right Handed	Male	27	187	92
08	Intact	Right Handed	Male	45	173	73
09	Intact	Right Handed	Male	23	172	63
10	Intact	Right Handed	Male	34	173	84
11	Intact	Right Handed	Female	32	150	54
12	Intact	Right Handed	Male	29	184	90
13	Intact	Left Handed	Male	30	182	70
14	Intact	Right Handed	Female	30	173	59
15	Intact	Right Handed	Male	30	169	58
16	Intact	Right Handed	Male	34	173	76
17	Intact	Right Handed	Male	29	175	70
18	Intact	Right Handed	Female	30	169	90
19	Intact	Right Handed	Female	31	158	52
20	Intact	Right Handed	Female	26	155	52
21	Intact	Right Handed	Male	32	170	75
22	Intact	Left Handed	Female	28	162	54
23	Intact	Right Handed	Male	25	170	66
24	Intact	Right Handed	Male	28	170	73
25	Intact	Left Handed	Male	31	168	70
26	Intact	Left Handed	Male	30	186	90

27	Intact	Right Handed	Male	29	170	65
28	Intact	Right Handed	Female	29	160	61
29	Intact	Right Handed	Male	27	171	64
30	Intact	Right Handed	Male	30	173	68
31	Intact	Right Handed	Male	29	185	98
32	Intact	Right Handed	Male	28	173	72
33	Intact	Right Handed	Male	25	183	71
34	Intact	Right Handed	Male	31	192	78
35	Intact	Right Handed	Female	24	170	52
36	Intact	Right Handed	Female	27	155	44
37	Intact	Right Handed	Male	34	190	105
38	Intact	Right Handed	Female	30	163	62
39	Intact	Right Handed	Male	31	183	96
40	Intact	Right Handed	Male	31	173	65

Fig. 3 NinaPro DB2 Subjects

This study will focus on the first 7 gestures, seen in fig 4. This choice is influenced by the fact that in [52] the researchers observed a performance gain when considering only a subset of the movements (Sri-iesaranusorn 2021).



Fig. 4 isometric hand configurations

Training will take place on subjects 1 - 20, and in-training validation will be performed on subject 40. Using all 40 subjects is beyond the CPU resources available for this study.

3.3.2 NinaPro Database 3

The NinaPro DB3 [7] contains the same gestures but from 11 amputated subjects. This is chosen as the non-ideal testing data due to Atzori et al.'s discussion in [53], in which it is noted that the parameters involved in regard to the amputation directly affect the model's

ability to classify the gestures. This variability in the data allows for more trustworthy and comprehensive results that help to reach the study's goals of solving the adaptability problem, as it must perform classification on unique signals from different patients in different environments.

Title	Hand	Laterality	Age	Height	Weight	Remaining Forearm (%)	Years passed by the amputation	Amputation cause	Phantom Limb Sensation Intensity	DASH Score	Use of cosmetic prosthesis (Years)	Use of kinematic prosthesis (Years)	Use of myoelectric prosthesis (Years)
01	Right Hand Amputated	Right Handed	32	172	86	50	13	Accident	2	1.67	0	0	13
02	Left Hand Amputated	Right Handed	35	183	81	70	6	Accident	5	15.18	6	0	0
03	Right Hand Amputated	Right Handed	50	178	82	30	5	Accident	2	22.5	0	8	8
04	Right Hand Amputated	Right Handed	34	166	68	40	1	Accident	1	86.67	0	0	0
05	Left Hand Amputated	Left Handed	67	175	75	90	1	Accident	2	11.67	0	0.4	0
06	Left Hand Amputated	Right Handed	32	172	66	40	13	Accident	4	37.5	0	12	0
07	Right Hand Amputated	Right Handed	35	185	75	0	7	Accident	0	31.67	0	0	6
08	Right Hand Amputated	Right Handed	33	175	80	50	5	Accident	2	33.33	0	0	4
09	Right Hand Amputated	Right Handed	44	180	95	90	14	Accident	5	3.33	0	0	14
10	Right Hand Amputated	Right Handed	59	177	86	50	2	Accident	5	11.67	0	1.66	0
11	Right Hand Amputated	Right Handed	45	183	75	90	5	Cancer	4	12.5	0	5	5

Fig. 5 NinaPro DB2 Subjects

3.4 Preprocessing

Preprocessing will be kept to a minimum to maintain simplicity and avoid introducing bias, however some essential steps will be taken.

3.4.1 Standardisation

Standardisation or normalisation is a vital step in preprocessing, and both are ways to ensure that all of the data is uniform. Vafaei et al. state in their comparative study [54] that normalisation is essential for decision-making methods due to the need for data to be aggregated into a single score per alternative (Vafaei et al. 2018). Such decision making is required of this experiment (e.g which gesture is the signal corresponding to?) Previous

research in classification has proven that normalisation has a positive influence on prediction accuracy [55].

This study will use an initial Standard Normal Distribution (SND) during preprocessing, via scikit's 'StandardScaler' function, to easily achieve uniformity within the data.

3.4.2 Window Sliding

Window sliding allows for time-series data to be segmented and passed into a CNN as individual training examples. Each window is a section of the time series data, and will have a corresponding gesture label. This implementation is required in order to process sEMG data with a CNN.

A window size of 200ms was picked according to the paper by Atzori et al. [7], and a stride of 20ms. These parameters can have a drastic effect on the results, as is discussed in [56]; finding the optimal window size takes time and testing. The 20ms stride is a starting point relative to the window size, and allows for a large overlap. It will be adjusted accordingly if errors occur, or the model doesn't behave as expected.

3.4.3 Reptile Tasks

For the Reptile model, as per the paper [11], tasks must be created from the training samples. Each task consists of a meta-task set, and a query set for training/testing.

For each subject's data, tasks are created, each with 5 examples. for each gesture performed. In this study, = 1000. A large number of tasks developed from the data allows for more diverse meta-learning, as the model has more to learn from.

The tasks' implementation will be explained in detail in section 3.6.1.

3.5 The CNN Model

In addition to the benefits discussed in the literature review, a CNN architecture has been chosen for this task due to its previous success in capturing local patterns and spatial dependencies in data [57], along with its success in sEMG classification [18], [19], [29], [58].

Another finding that contributed towards the choice of a CNN is that they are invariant to small translations in their input data, as shown by Myburgh et al. [59]. This is an important feature for this study's task in particular, as it means that ACEnet will be able to recognise patterns regardless of any shifts in electrode placement during the recording process, allowing it to be robust towards the slight variations in complex sEMG signals.

The Reptile and Standard architectures both use identical network models. As seen in Fig. 6, it comprises three 'modules', two identical convolution modules and an output module.

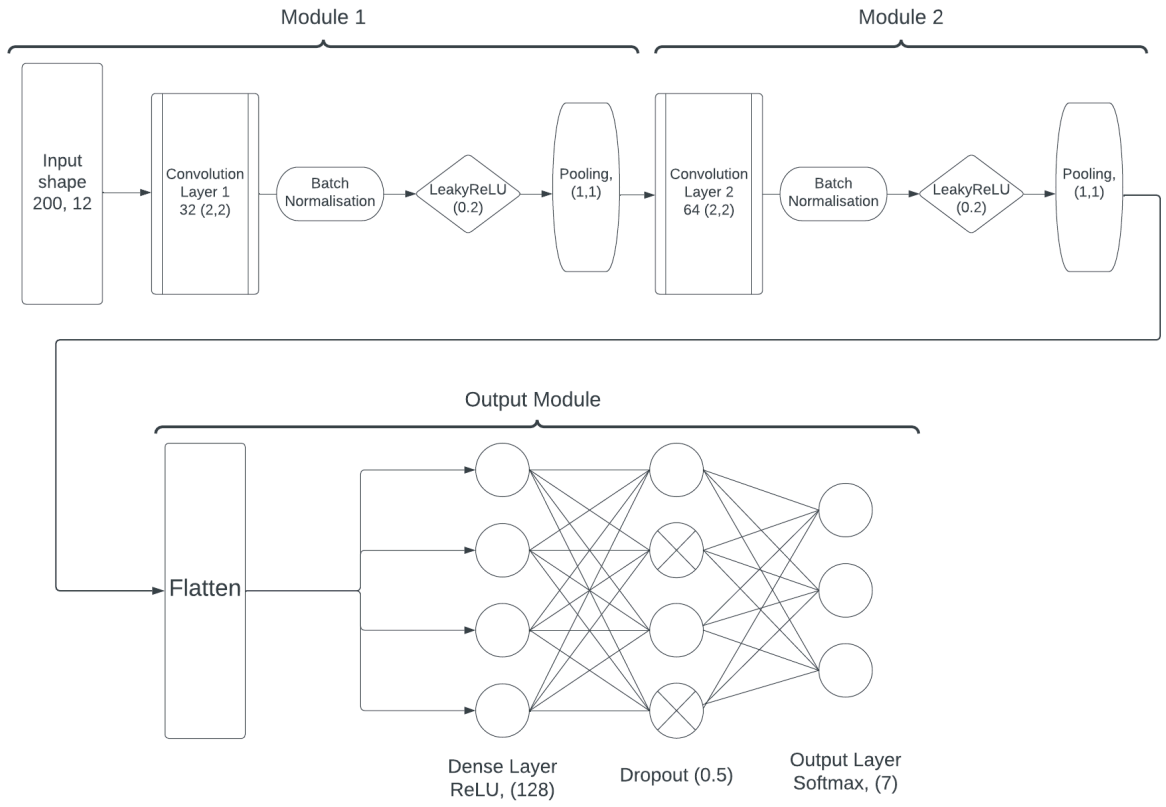


Fig. 6 CNN Architecture Diagram for both Reptile & Standard variants

3.5.1 Convolution Layers

The convolution layer is of size 200 * 12, matching the data's window size * electrodes. This layer captures the patterns and spatial data, and is the backbone of the model.

3.5.2 Batch Normalisation

Performing batch normalisation speeds up the training process by reducing the internal covariate shift [60], which is when the distribution of each layer's inputs change, slowing down training (Ioffe, Szegedy 2015). As MAML based models are computationally expensive to train [61], it's vital to take necessary steps to improve performance where applicable, which makes batch normalisation an important part of the reptile model's architecture.

3.5.3 LeakyReLU Activation

A standard ReLU is often used in deep learning models, and has shown itself to enhance network performance when tested by LeCun et al. [62] (the pioneer of CNN's). It also solves issues regarding vanishing gradients [63]. In further research, the ReLU continues to perform well in terms of accuracy, however it is outperformed by adapted versions [64] [65]. The ReLU also comes with a unique issue; the 'Dying ReLU', which is described by Lu et al. [66] to be problematic for complex data where preservation is vital, such as sEMG data.

The 'LeakyReLU' [67] combats the dying ReLU issue by allowing for non-zero outputs [68]. This allows for the retention of potentially defining information.

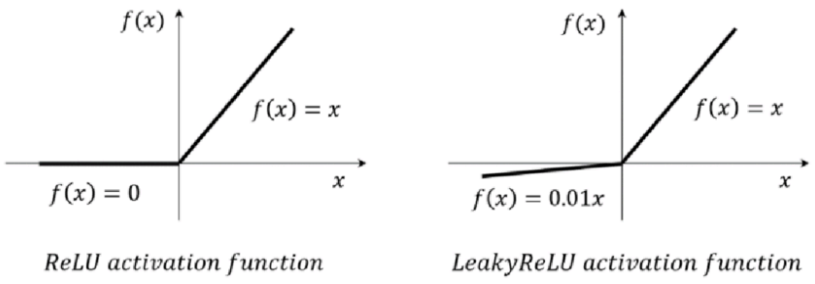


Fig. 7 - Comparison of ReLU and LReLU [69]

Due to the nature of the data used in this study being complex, it is vital that we preserve as many defining features as possible, therefore the LeakyReLU will be implemented as the activation function.

3.5.4 Max Pooling

Pooling is introduced with a size of 2,2 to reduce the computational complexity of the model, leading to faster convergence and improved generalisation performance [70, p343], [71].

3.5.5 Dropout

Dropout is applied as a setting of 0.5 (50%). This is another essential regularisation technique to aid in overfitting prevention [74] and is frequently implemented in previous CNN studies [73] [72].

3.6 Training

The hyperparameters for this study's Meta-Learning will mainly mimic those of [38], as it has the most similar goals to this experiment.

3.6.1 Reptile Meta-Learning Loop

The model will use each meta task's 5 examples to learn from, simulating the few-shot scenario that would apply in a real world scenario. The training takes place in a 'meta (inner) loop', and the model updates take place in the 'outer loop'.

Each outer loop iteration creates a copy of the current ACEnet model named the task-model. Then, in the inner loop, It's trained on the meta-task, and tests it on the query set.

Back in the outer loop, the task-model's learned parameters are saved to a list.
The average parameters of each set in the list are then calculated by dividing the summed parameters by the number of them present.

Finally the 'Reptile sum formula' seen below is the update step for the main ACEnet model's parameters, as described in the pseudocode from [30]. An additional 'learning rate' is

included as simply to be used for the calculation, and is set to a basic rate of 0.1, also taken from [30].

$$Update: \Phi \leftarrow \Phi + \varepsilon(W - \Phi)$$

Φ = *initial parameters vector*

W = *final parameters learned on the task*

ε = *epsilon, the step size (0.1)*

This process is then repeated for outer loop iterations. In this study, the training will end when it begins to converge and produce diminishing returns, thus .

A snapshot of the ACEnet model will be taken at each iteration, and the one that achieves the highest accuracy on the validation test data will be saved and used as the final model.

3.6.2 Optimisation

For this experiment, the 'Adam Optimiser' is used in place of Stochastic Gradient Descent, as it has been proven to be successful for computer vision tasks through previous literature [75], [76] and clearly works well with the methods of this experiment as it is also employed in [38].

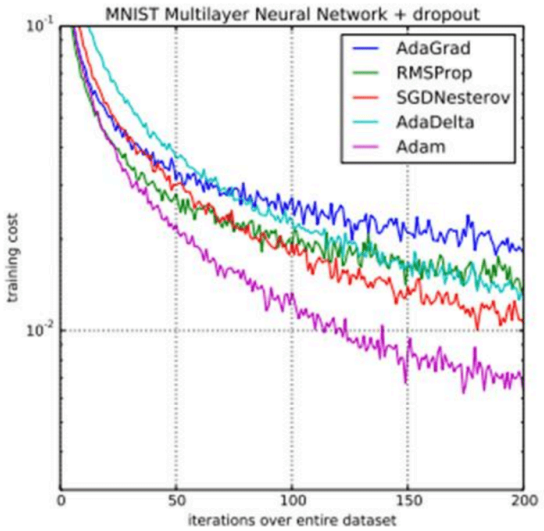


Fig. 8 Adam outperforming other optimisers [75]

3.6.3 Loss Function

'Categorical Cross Entropy' is used as the loss function, as it succeeds with multiclass problems due to its ability to minimise the distance between the predicted and correct class, providing an accurate probability [77].

3.6.4 Meta Loop Hyperparameters

The batch size and learning rate have been kept low to ensure that the network generalises well and avoids overfitting [78].
The meta learning rate is set to 0.1, as this encourages the model to converge faster [79].

The model also has a meta batch size of 8, which means it will sample 8 tasks at each iteration to learn from; Fan et al. [38] state "Overall, generalization and adaptability are better when meta batch size is set to 8".

Finally for the meta loop, = 1, meaning there is 1 step of ADAM optimisation performed on each task. Performing just a single update to the parameters encourages the model to learn rapid adaptability, as is stated by Fan et al. in [38]

3.6.5 Outer Loop Hyperparameters

The outer loop has a learning rate of 0.001, and a batch size of 32. These are the default parameters set with the Tensorflow ADAM implementation, and they can be seen performing well in the literature on similar model architectures [80], [81, p600]. These will be adjusted if required, according to the results.

The comparison model (the standard CNN) will copy the above settings to ensure a fair test is conducted.

3.7 Testing

The project is split into two major sections*: training* and *few-shot testing*.

3.7.1 In-Training Testing

During the training phase, each model is also validated at each iteration on subject 40, which contains unseen data. This allows for the comparison of validation accuracy between the models, and acts as a general indication of the performance at each step.

3.7.2 Few Shot Testing Environment

This study's few-shot testing phase will take the form of 7-way 5-shot. This means the pre-trained models are shown 5 training examples for each subject from the NinaPro DB3 [7], which contains the sEMG data of 11 amputees, then tested on the rest of the subject's gestures, as it would be in real life.

3.8 Data Augmentation

Window Warping (WW) also known as Dynamic Time Warping, will be the augmentation method used to expand the dataset, as recommended by Iwana et al. [13]. The evidence from [13] that WW is one of the best overall choices is irrefutable, as it is compared with such a vast number of other methods.

Window warping takes a section of the time-series data, and distorts it as seen in fig 9 by interpolating the data points in the chosen window, and introducing new ones.

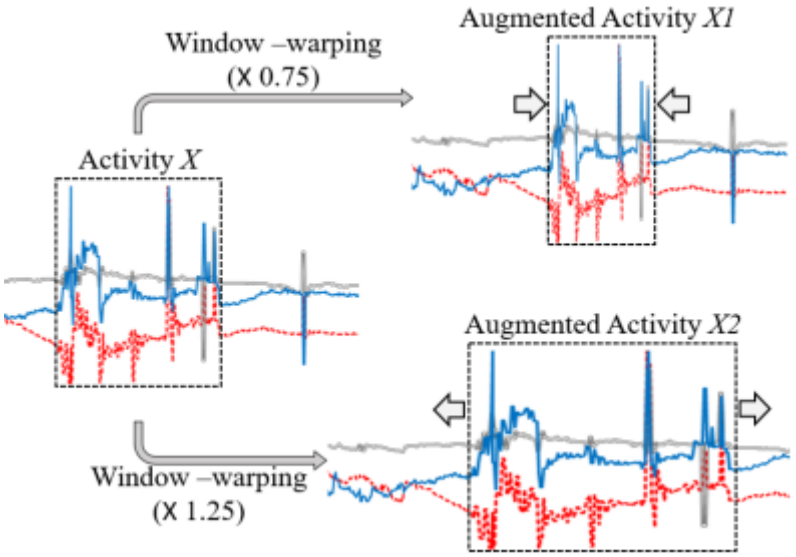


Fig. 9 - Window warping demonstration from [82]

In a further study by Dau et al. [83] the researchers utilised WW to generate more synthetic samples. They also found the warping factor parameter to be absolutely crucial to success, and without it being set correctly, it is impossible to produce high-quality results [83] (Dau et al. 2018). The warping factor will therefore be copied from [82] as the study's settings appeared to improve model performance substantially, seeing results in a 51% and 16.2% increase in F-1 score.
A window size of 30 was chosen as an arbitrary number, and may be adjusted accordingly if required.

This use of WW to expand the dataset will be copied in this study, as it proved to be successful for Dau et al [83]. Furthermore, Guennec et al. [84] the original proposers of WW, used the augmentation method to deal with small datasets, which proves that it successfully answers this study's aims of dealing with small datasets.

3.9 Analysis

Matplotlib will be used to plot the metrics as line graphs, allowing for the identification of the convergence point and highest accuracy.
The x axis will represent the iteration, and the y axis will represent the classification accuracy.

If overfitting is identified by the comparison of the above metrics, the learning rate will be adjusted to find the sweet spot, along with the batch size, window size for window sliding, and window size for WW.

Both models will have a set of analysis from the augmented and non-augmented experiments. These will all be compared in a table.

3.9.1 Reptile Task Metrics

Firstly, the validation data accuracy and the training accuracy of each individual Reptile task will be plotted in succession on the same graph, to see if the tasks are being learned from successfully.

Secondly, the validation loss and training loss will be visualised similarly, to check that they are also producing normal results, and hopefully decreasing over time.

3.9.2 ACEnet Model Metrics

Only the validation accuracy & loss of subject 40 will be used for the main ACEnet model, plotting each iteration of the outer loop. This is the only important metric for this part of the study, as it indicates the model's overall success at classifying unseen data. Again, these metrics will be visualised on a line plot to see how the model acts over time.

The standard model metrics will follow the reptile task metrics format, comparing the training and validation accuracies on a line plot.

3.10 Ethical Considerations

This study has undergone an ethics report and has been accepted, but some concerns must be highlighted.

The clearest concern is the privacy of the subject's date used in the project, however the subjects within the study will not be explicitly referred to and their data is anonymised. All of

the subjects consented to the usage of their data, as well as it being cleared for research use in the originating paper [7].

Another concern is bias being introduced, which has been addressed in the sections above by ensuring that both models run with the same parameters, the datasets are the same, and minimal additional methods/tools are implemented to keep the focus on the adaptability comparison of the raw models.

This leads to a vital concern; that the results must be dependable, as applying an inaccurate model to a real world scenario could present dangerous outcomes. As previously mentioned, this study is intended to provide the basis for further research, rather than providing an end solution.

4.0 Experiments & Results

Below are the results for both the standard and ACEnet models. The first phase is with the normal dataset, and the second phase uses the augmented dataset.

4.1 Phase I: Unaugmented Experiment

4.1.1 Standard Model Results

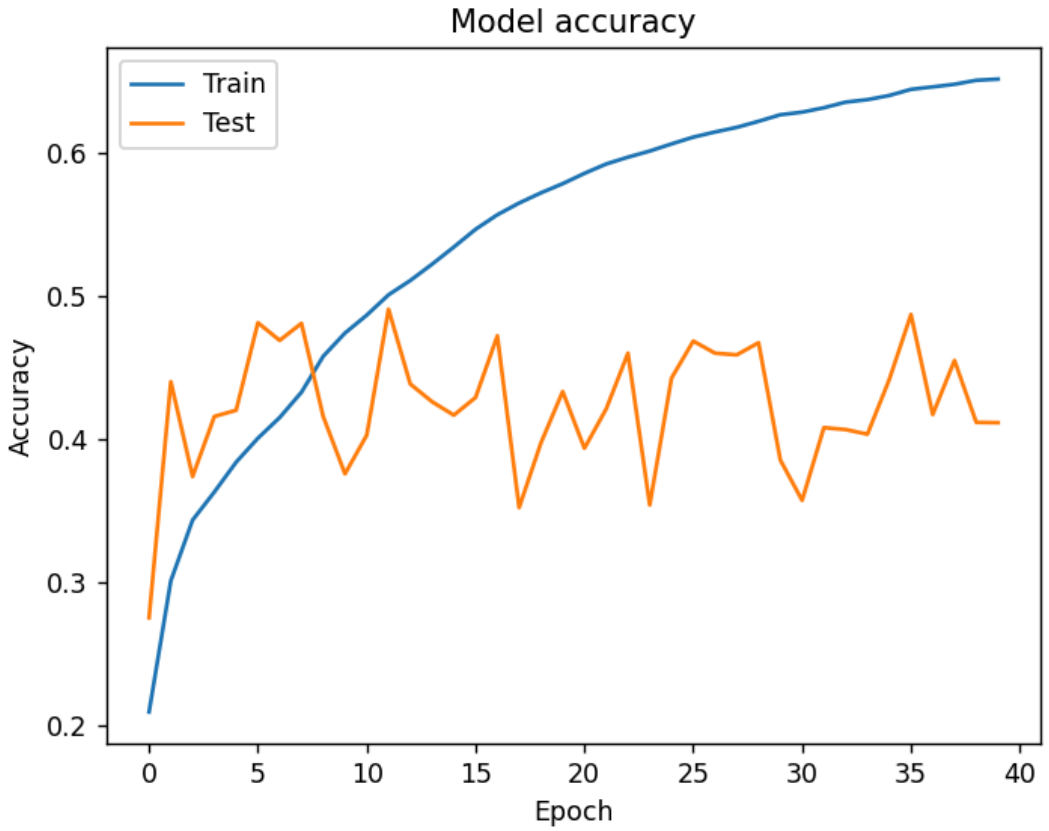


Fig 10. The standard model's training and validation accuracy over time

The training accuracy increases as the model learns over time, converging at **40** epochs with an accuracy of around **0.65%**; a normal result for complex data, similar to that of [38]. The validation accuracy is erratic, and appears to immediately achieve a max accuracy of **0.49%**

This implies that the model is simply guessing, and it cannot predict well on the validation subject, based on the training data it has received. This is the expected output.

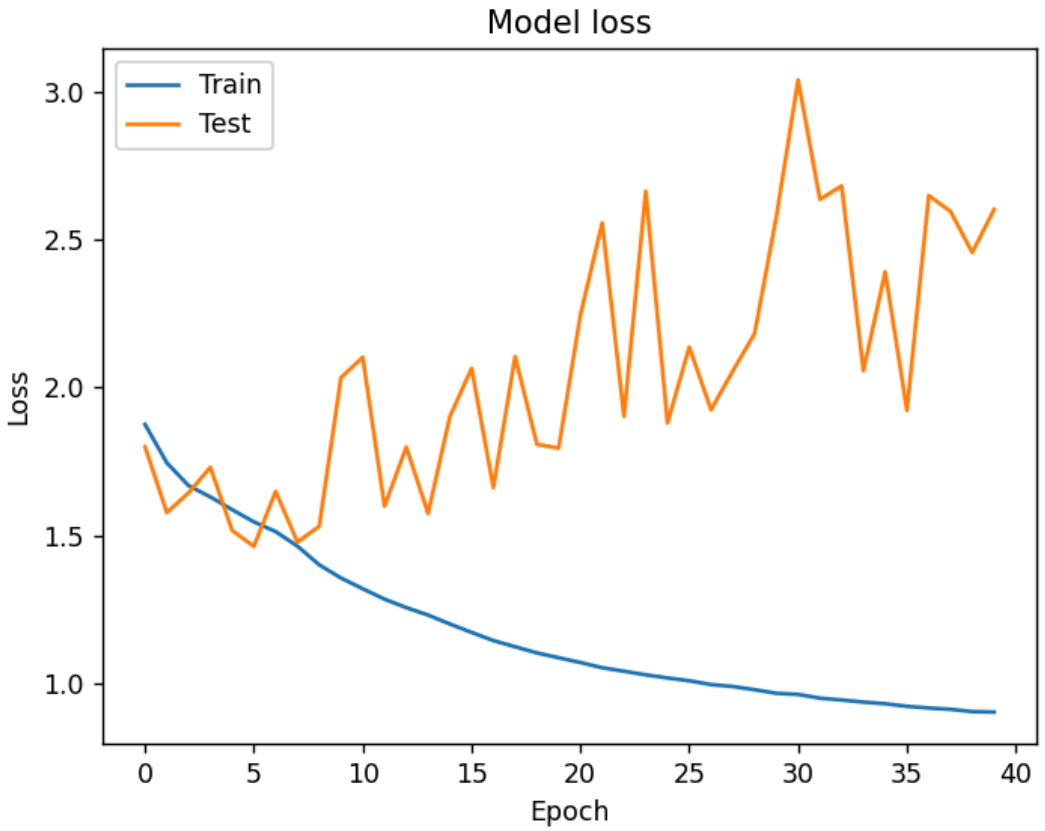


Fig 11. The standard model's training and validation loss over time

The standard model achieved a max validation accuracy of **0.48** and a smallest validation loss of 1.5.

The validation loss increases over time, indicating that it is not learning from the training data. Again, this is the expected output, as the unseen data has no correlation to the training data, and is the problem that ACEnet aims to solve.

4.1.2 Reptile Model Results

Fig's 12 and *13* represent all of the individual task iterations combined together. Fig. 16 represents the ACEnet overall model's accuracy on the unseen validation data (subject 40).

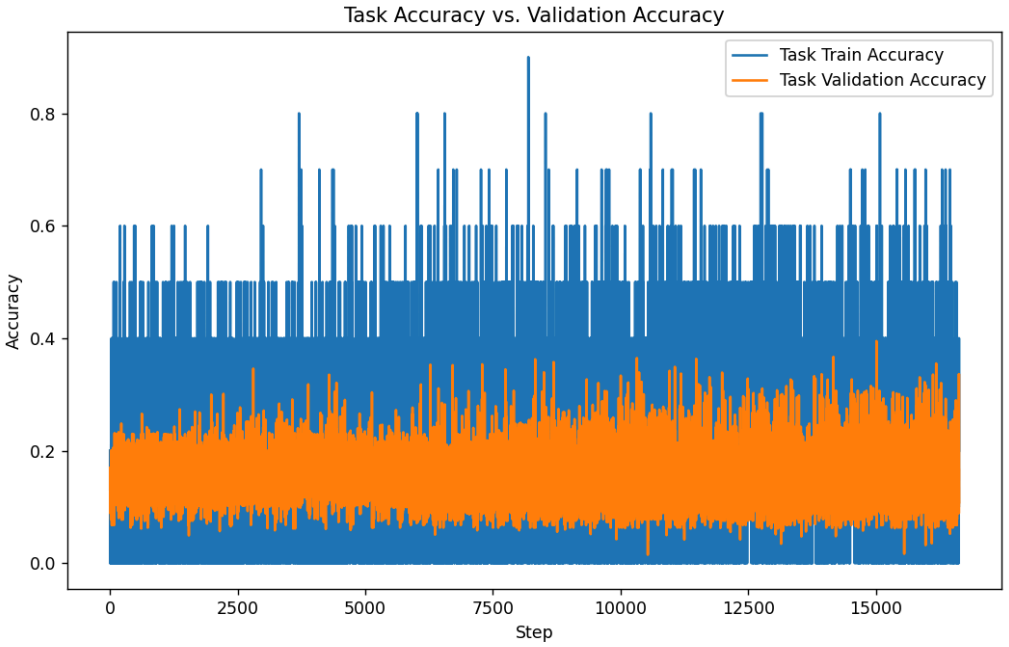


Fig 12. ACEnet's inner task training accuracy & validation accuracy over time

The model performs as expected; each new set of tasks requires new learning, as there is only one step of SGD performed, therefore displaying some failures and some success. Over time the model learns from the previous tasks, which can be seen as the white space decreases, and more accurate decisions are made.

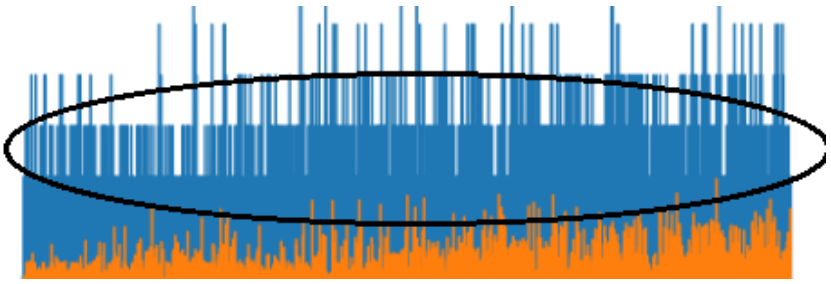


Fig 13. The tasks beginning to have more accurate classifications over time

The task validation accuracy also begins to increase over time, implying that the range of tasks being learned from is increasing the model's adaptiveness to unseen data. However, it does converge at around 9000 tasks, meaning the improvement is minor.

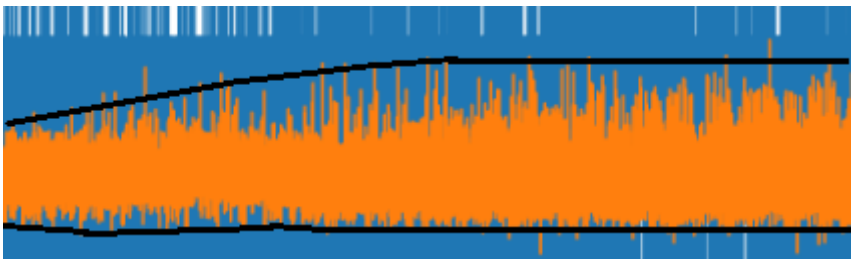


Fig 14. The validation accuracy increasing and converging

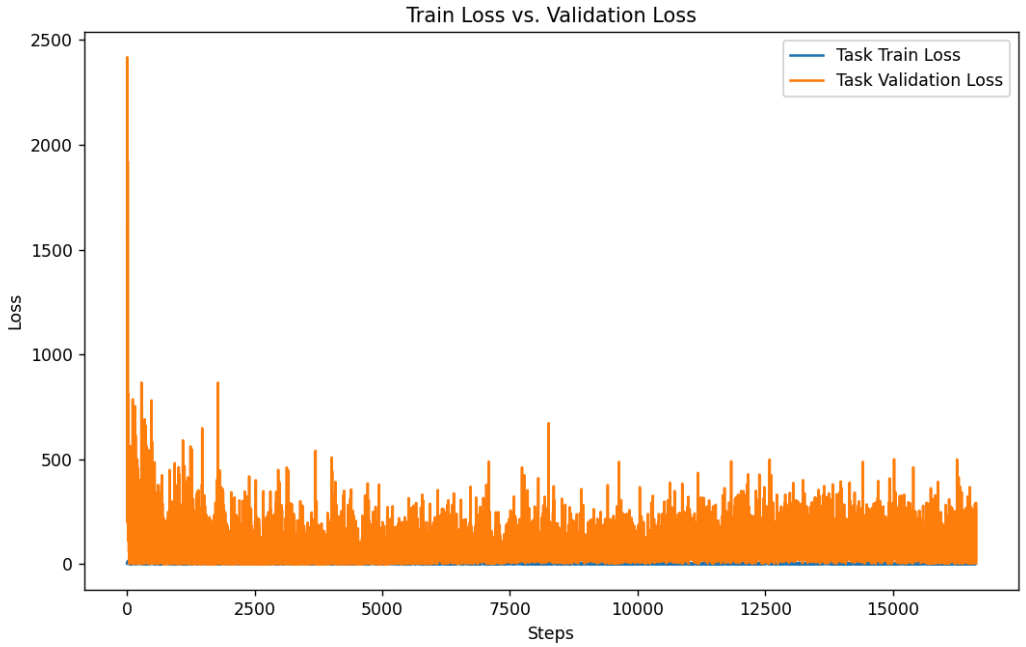


Fig 15. ACEnet's inner task training loss & validation loss over time

The training loss is heavily overshadowed by the validation loss. It appears to start increasing again around 6000 tasks, implying that the model is beginning to overfit to specific tasks; likely those that it has completed before, as they are selected at random.

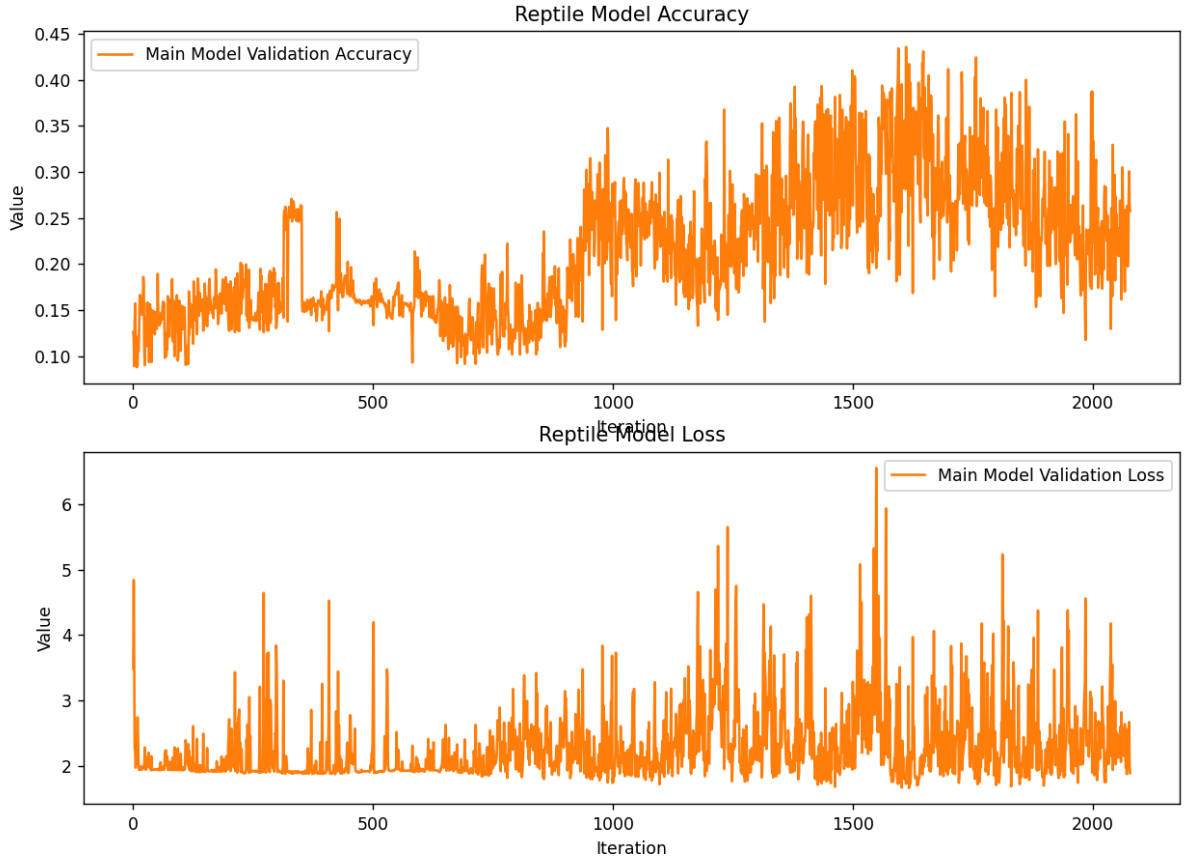


Fig 16. ACEnet's overall model validation accuracy & validation loss over time

Moving on to the main model, ACEnet converged at **1650** iterations and began to decrease from there on, at which point the best model was taken. It achieved **0.42** accuracy.

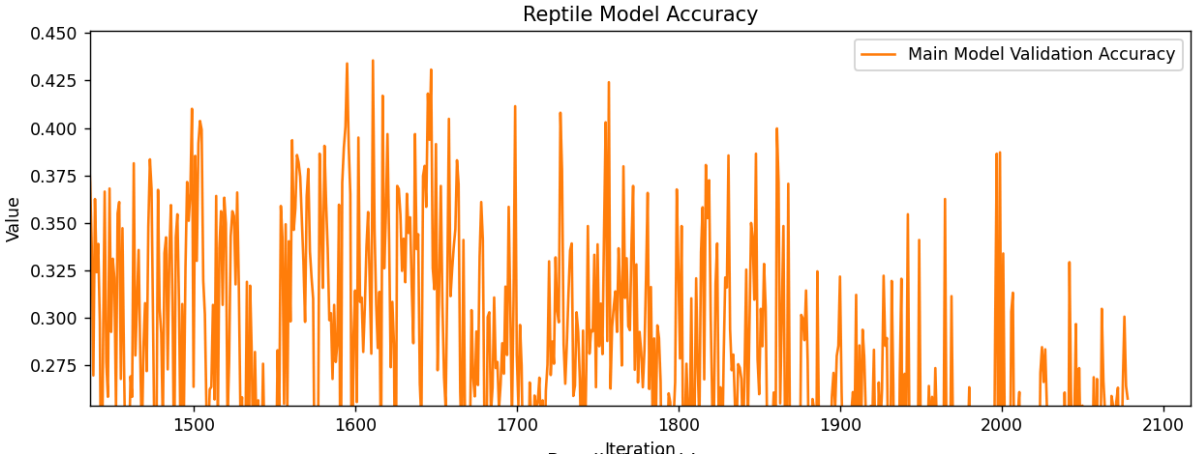


Fig 17. Zoomed in accuracy at convergence point

The validation loss becomes more and more erratic, indicating signs of overfitting and dependency on the training data.

Both models ran until they converged, with the Reptile model performing far more iterations due to only one update being performed each time.

The models achieved a similar accuracy to each other. The standard model seemed to jump straight to the 0.40-0.50 range, and proceeded to drop down and then increase again with each iteration, signalling that it is making educated guesses.

At this point, it is difficult to say if the primary research question has been answered, as ACEnet did not produce a higher validation accuracy than the standard model. It did however appear to learn from the unseen data over time, as it gradually increases and eventually reaches its peak, at which point it can't learn any new defining features, indicating that it *may* be more adaptable. Phase II and the Few-Shot Testing will solidify the answer.

4.1.3 Few-Shot Testing on Unseen Data Results

Table 1: Results from 7-way-5-shot testing on unseen data

Subjects	Reptile Model Accuracy	Standard Model Accuracy
Subject 1	**0.1737777143716812**	0.11593826860189438
Subject 2	0.19353775680065155	**0.2103527933359146**
Subject 3	**0.17877094447612762**	0.11894785612821579
Subject 4	**0.16975508630275726**	0.167292982339859
Subject 5	0.10472226142883301	**0.14062702655792236**
Subject 6	0.13139158487319946	**0.14420711994171143**
Subject 7	0.14137794077396393	**0.15093588829040527**
Subject 8	**0.35062485933303833**	0.2895543575286865

Subject 9	**0.2674501836299896**	0.17858678102493286
Subject 10	**0.15144199132919312**	0.13843058049678802
Subject 11	0.19438840448856354	**0.27947917580604553**

Reptile achieved higher accuracy than the standard CNN model on **6/11** occasions, a fairly evenly matched turnout, and therefore the hypothesis is a failure.

4.2 Phase II: Augmented Data Experiment

In phase II, the augmented dataset is used for training. It contains double the amount of training examples as the original dataset.

4.2.1 Augmented Standard Model Results

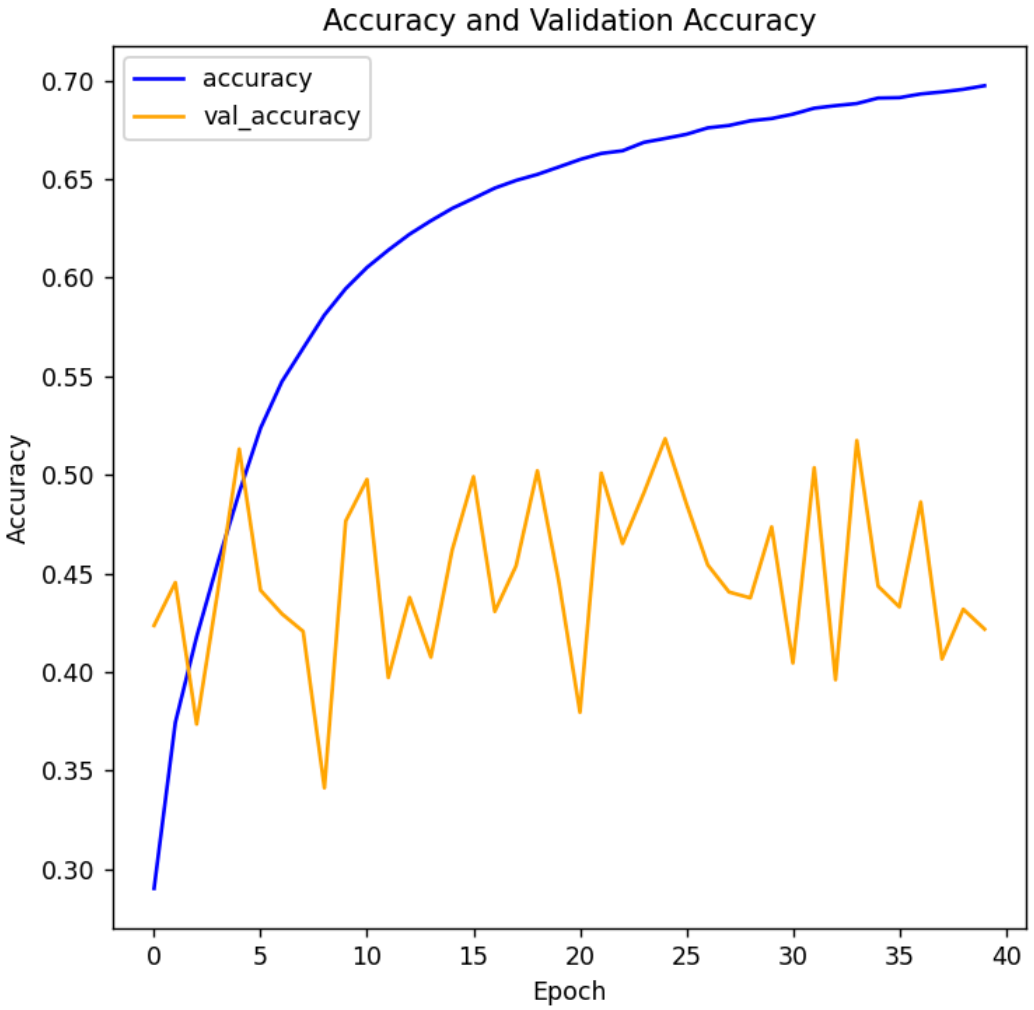


Fig 18. The augmented standard model's accuracy and validation accuracy

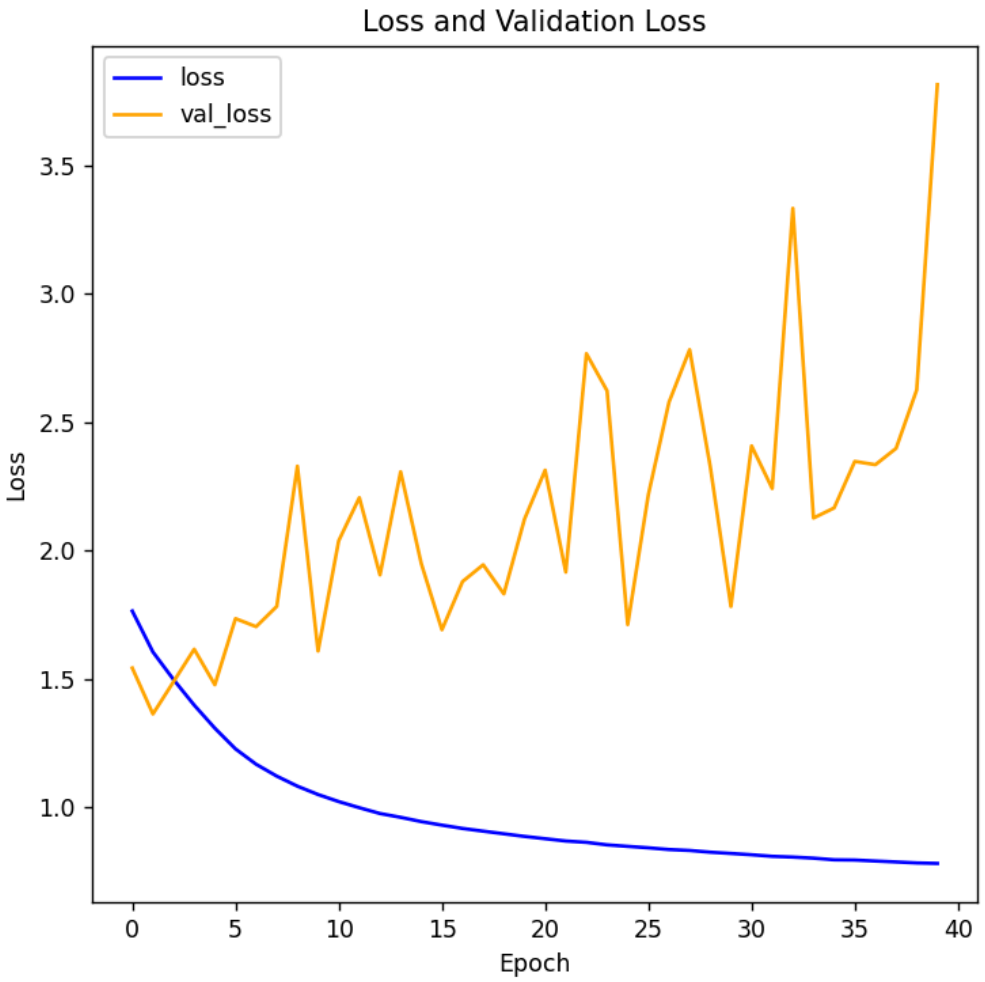


Fig 19. The augmented standard model's loss and validation loss

Fig's 18 & 19 show that the augmented dataset achieved slightly better results, as it did manage to learn faster, producing sharper learning curves, and a higher max validation accuracy of **0.52**, which answers the secondary research question based on augmentation satisfactorily. It also achieved a final training accuracy of **0.70%**, as opposed to the unaugmented model's **0.65%**, backing both research questions in terms of accuracy.

4.2.2 Augmented Reptile Model Results

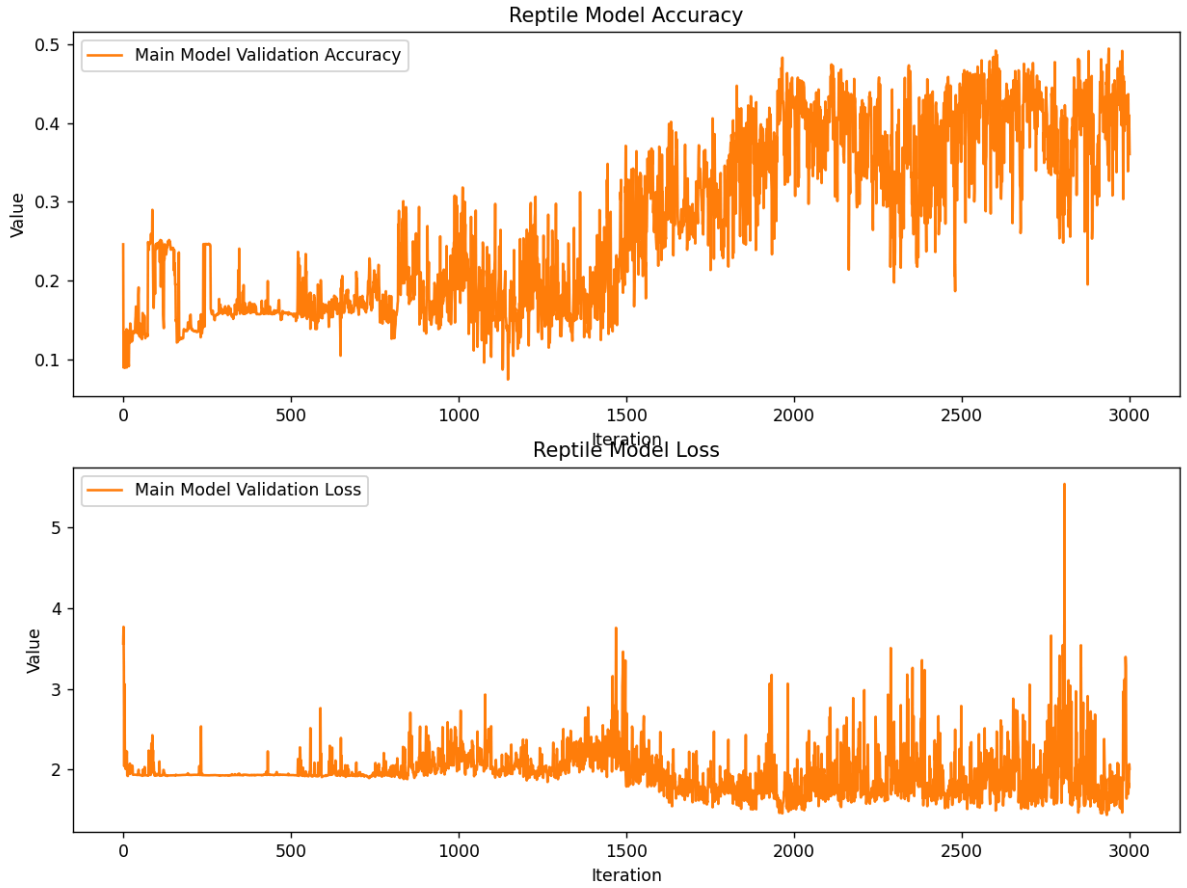


Fig 20. Augmented ACEnet's overall model validation accuracy & validation loss over time

As can be seen in *fig 20*, ACEnet surpasses its unaugmented predecessor; it begins to converge later at **2000 epochs** instead of **1600**, and even still after this, continues to make some very slight increases as seen in *fig 21* and *fig 22*.

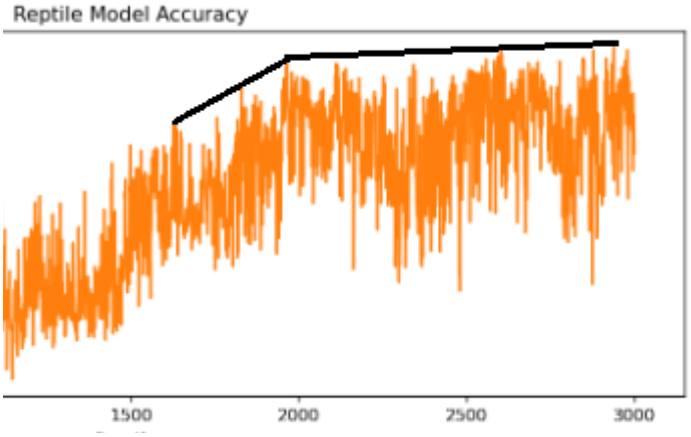


Fig. 21 Augmented ACEnet's accuracy outperforming Phase I's non-augmented training

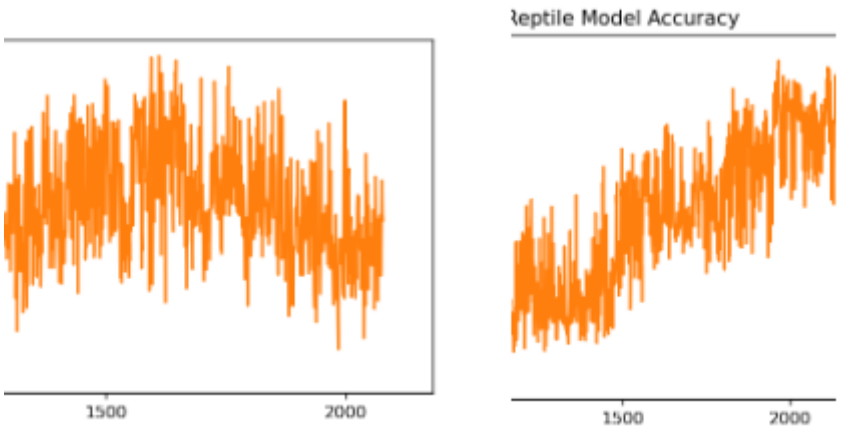


Fig. 22 Unaugmented ACEnet (left) declining between 1500 - 2000 epochs whilst Augmented ACEnet (right) continues to improve in the same region

Overall the augmented run achieved a total max validation accuracy of **0.49** in comparison to the phase I's **0.42**, again answering the research question on augmentation producing a more accurate result.

4.2.3 Augmented Few-Shot Testing on Unseen Data Results

Table 2: Results from 7-way-5-shot testing on unseen data when used with augmented ACEnet model

Subjects	Augmented Reptile Model Accuracy	Standard Reptile Model Accuracy
Subject 1	0.12255220115184784	**0.16716378927230835**
Subject 2	0.12611275911331177	**0.17985492944717407**
Subject 3	0.13244879245758057	**0.19110800325870514**
Subject 4	0.15718543529510498	**0.17027342319488525**
Subject 5	**0.12527644634246826**	0.1189020425081253
Subject 6	**0.14420711994171143**	0.14058253169059753

Subject 7	**0.1446966677904129**	0.14137794077396393
Subject 8	0.24428200721740723	**0.34826692938804626**
Subject 9	0.22172853350639343	**0.2697434425354004**
Subject 10	**0.14942991733551025**	0.12488263100385666
Subject 11	**0.20392444729804993**	0.19622226059436798

The augmented reptile model achieved higher accuracy than the standard Reptile model on **5/11** occasions, once again showing negligible results; another failure when performing on new data in a few-shot environment.

4.3 Unseen Data Accuracy Results Table

This table reflects each model's ability to adapt to unseen data of a similar structure, as each accuracy result was achieved by testing on Subject 40's data.

Table 3: The highest accuracy achieved on Subject 40's Unseen Data

Model	Phase I (Unaugmented)	Phase II (Augmented)
ACEnet	0.42	0.49
Standard Model	0.49	0.52

From the combined results in tables 1 and 2, it's clear that implementing Reptile did *not* increase the adaptability of the model in a Few-Shot environment, answering the primary research question negatively.

Table 3 shows that augmentation *does* increase the overall accuracy, and showed a greater improvement when used with ACEnet compared to when used with the standard CNN, answering the secondary research question positively.

The standard model still outperformed ACEnet in terms of accuracy overall, and therefore the hypothesis has been disproven.

5.0 Critical Analysis & Discussion

5.1 Strengths

The primary strength of the study is its contribution to the literature gap; an entirely new avenue was explored, and the gap has now been successfully filled; Reptile was successfully tested within the field of sEMG classification. Expanding upon this, the study has laid the groundwork for future research that can combat the issues encountered in this study, and hopefully make good use of the results to ultimately improve the current state of bionic prosthetics in real world scenarios.

Another strength of the study is that the aims and objectives were achieved. Creating ACEnet was the primary aim, and this was achieved successfully through the proposed tools and methods. The two sEMG datasets were also successfully used for training and validation, despite not yielding the expected results.

Furthermore, the successful implementation of Window Warping provides an insight into improving ACEnet's performance in future studies, as well as other MAML models.

Overall, the study has provided a valid contribution to a niche field, and filled a literature gap, delivering more information and results around an unexplored topic. With the insight provided, future research can adapt this study's methods to produce improved results.

5.2 Limitations

The main limitation of the study is the lack of preprocessing, which is evidently an important part of the success in previous studies [18], [19], [38].

The small dataset was also a limitation as previously discussed; a point reinforced through Phase II's expansion showing improved results. Despite this, it should be taken into

consideration for further research that having a large number of synthetic samples could potentially train the model on unrealistic data.

The model was also only compared with another CNN. It would have been more comprehensive to also test the dataset with other well known models in the sEMG field, such as RNN's, or even an RNN-CNN hybrid.

5.3 Comparison to Previous Literature Results

5.3.1 CSAC-net - Few-Shot Accuracy Results

Fan et al. [38]'s best few shot classification scores are greater than this study, likely due to their inclusion of STFT preprocessing.

Dataset	1Shot-Top1	1Shot-Top5	5Shot-Top1	5Shot-Top5
DB-a	72.50	65.00	77.50	76.49
DB-b	62.50	57.50	83.00	81.00
DB-c	80.00	70.00	83.50	80.99

Fig. 23 Highest few shot testing from Fan et al. [38]

As seen in Table 2, ACEnet achieved a max of 0.24, in comparison to Fan et al [38]'s 0.83 in *fig. 23*, a huge difference, indicating that correct preprocessing of sEMG signals is vital to the model's comprehension of the data.

Fig 24. Is one of Fan et al. [38]'s ablation studies, which tests raw sEMG data in a few-shot test, and produces similar best results to this study (0.34 & 0.35 in Tables 1 & 2), confirming that preprocessing is vital.

input	Dataset	1Shot-Top1	1Shot-Top5	5Shot-Top1	5Shot-Top5
Raw sEMG	DB-a	32.50	30.00	45.00	44.00
	DB-b	30.00	27.50	40.00	39.89
	DB-c	37.50	35.00	45.00	44.00

Fig. 24 Fan et al. [38]'s ablation study for raw sEMG data

5.3.2 sEMG-Based Gesture Recognition with Convolution Neural Networks - Training Accuracy Results

In this paper, Ding et al. [19] Use raw sEMG signals with no preprocessing, and pass it to a CNN, which is similar to this study. Additionally they also use the NPDB2. however their experimental setup and architecture is very different.

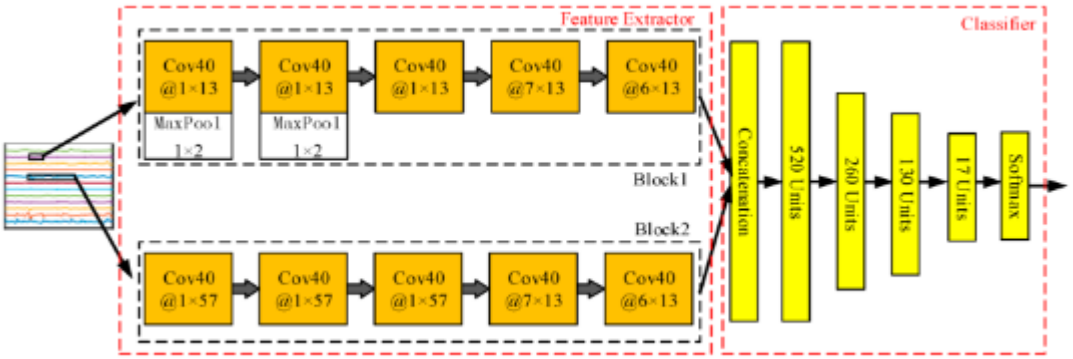


Fig. 25 Ding et al. [19]'s CNN architecture C-B1PB2

The above architecture uses multiple modules, namely a feature extractor and a classifier to achieve its results, making it somewhat incomparable. It achieved slightly better training accuracy results than this study.

Table 3. Performance comparison with state of the art methods on all hand gesture of DB2.

Methods	Atzori et al. [35]		Weidong et al. [36]			Xiaolong et al. [39]		Proposed
	CNN	RF	CNN (Single Frame)	CNN (20 Frames)	KNN	CNN	SVM	CNN
Accuracy	60.27%	75.27%	76.1%	77.8%	70%	78.71%	77.44%	78.86%

Fig. 26 Comparison chart from [19] depicting the training accuracies achieved across various similar studies

This study achieved a training accuracy of 65% in phase I and 70% in phase II, which is similar to the results of the compared studies in *fig 26,* showing that ACEnet is performing correctly, and the study's results can be trusted.

5.4 Conclusion

The un-augmented Reptile model and the standard CNN seemed to perform similarly. However, the augmentation of training data to expand the dataset has a positive impact on the accuracy and adaptability of the model, answering the secondary research question, and proving that meta-learning based models do perform better on larger, more diverse datasets, such as Omniglot. This shows that the dataset used in the non-augmented phase is simply too small for ACEnet to learn enough from to provide accuracy classifications.

It is also notable that the standard CNN model only performed slightly better on the augmented data, whereas ACEnet saw a much larger increase, indicating that ACEnet's ability to learn from a diverse dataset may be better than the standard CNN model.

Despite the success in the area of augmentation, the primary hypothesis failed, and the primary research question was answered negatively. ACEnet didn't perform better than the

standard trained CNN model in terms of accuracy, in both validation testing and the few-shot testing on amputated subjects.

The findings indicate that raw sEMG data is simply too complex to be learned from with just a few examples, especially when the test data is significantly different to its training data.

Overall for the sEMG field, despite the hypothesis not being supported, these findings highlight the challenges surrounding learning from small, complex data. Meta Learning still shows promise in tackling this as explored in the literature, however the methods and preprocessing techniques used must be refined. The study confirmed that preprocessing steps are vital to accurate classification of sEMG signals, and that STFT specifically helps a lot.

A research gap has also been identified and filled, which is a step forward in the field, and progress towards opening new possibilities for real world applications of bionic prosthesis, as well as laying the foundations for future research. Multiple issues were addressed, which is helpful for implementations across the board.

This study sparks a debate as to whether Reptile is useful *at all* for classifying sEMG signals, and it would be beneficial to test it in conjunction with different methods to verify or falsify this.

5.5 Future Work

In future research based on ACEnet, as explored in the literature review, preprocessing steps such as Short-Term-Fourier and Fast-Fourier have shown improved accuracies when implemented [18], [19], [38], [85] passing the data as spectrograms and increasing feature detection, and should be considered for future research that builds upon this study beyond just the model's adaptability.

As mentioned in the methods section, playing with the parameters such as the window size and step could provide better results. Ding et al. [19] state that the length of a window is the compromise between time latency and classification accuracy, therefore finding the sweet spot is important.

Another consideration for future work is the use of an RNN, as opposed to a CNN. RNN's can retain temporal dependencies in the data, as discussed in [23] and [21], and as a result of their architecture, do not require window sliding as preprocessing as they can process time series data directly, thus cutting out an entire step [86]. Bittibssi et al. [85] boast a near 99.6% prediction accuracy, proving RNN's exceptional performance. For future studies, an RNN version of ACEnet could be trialled and compared to the results of this study's model.

Providing a more diverse dataset through Window Warping would be beneficial according to this study's results. It would also help to train on an amputee sEMG dataset to begin with, as this would translate well to real life implementations due to the similarity in data.

This study chose to start with very harsh conditions, to provide valid results for future work on ACEnet, and signal areas for improvement. In future research, the conditions could be relaxed and closer to the standard real life implementation, to allow for a different approach that may welcome unique results that are unachievable through a harsher experiment.

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